# BA820 – Project M2

## Cover Page

- **Project Title:** Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Inchara Ashok

## 1. Refined Problem Statement & Focus

The domain question for this milestone was: "*How has the "typical" age gap in on-screen romantic couples shifted across decades (1930s–2020s), and where do the most significant inflection points occur?"* We initially expected a simple downward trend over time. Instead, the data showed that the "average" age gap is misleading, as it masks two distinct behaviors occurring simultaneously, something a single timeline cannot capture.

Using K-Means clustering, I found that Hollywood history is not a smooth progression but is divided into distinct casting regimes. The modern era is not merely a smaller version of the Golden Age; it is mathematically distinct. While contemporary casting is relatively stable, the 1960s and 1980s appear as turbulent transition periods in which legacy industry practices collided with changing social norms. A key insight emerged when separating Couple 1 from other couples: large age gaps could suggest star bias, with the most prominent actors paired with much younger partners in ways rarely seen among the film's remaining cast. Testing the data against the "half-your-age-plus-seven" rule further showed that although social norms have tightened, some films continue to violate them. Silhouette analysis highlighted these outliers, such as 1980s James Bond films which function as cultural fossils preserving Golden Age casting norms long after their time. This suggests that power dynamics and franchise legacy can be as influential as historical period in shaping casting decisions.

In conclusion, I reframed the research question because "How has it shifted?" captures only surface-level trends. Asking "**To what extent do latent industry power dynamics and evolving social norms define the transition between distinct casting regimes in Hollywood history?**" focuses instead on the mechanisms driving change. This shift moves the analysis beyond description to explanation, and accounts for how films like 'Moonraker' and 'Raiders of the Lost Ark' can coexist within the same decade, something a simple timeline average could never explain.

## 2. EDA & Preprocessing: Updates

Early analysis showed that age gaps do not change gradually over time but instead shift through sharp inflection points. The most significant transition occurred mid-century, when the extremely high median gaps of the Studio System era dropped abruptly, followed by more modest variation in recent decades. This demonstrates that the "typical" age gap is not a fixed industry standard but a historically sensitive phenomenon. Because the data is heavily imbalanced across periods, decade-level comparisons provide a more accurate view of these casting changes than overall averages.

A major methodological shift involved moving from treating age as a raw value to evaluating it against social and industrial benchmarks. I introduced a "Rule of Seven" threshold for each couple, since plotting age gaps alone could not distinguish between socially typical and excessive pairings. By creating a binary indicator for rule violations, I transformed the raw data into a narrative of evolving social norms, revealing that the mid-century decline reflects a measurable move toward more socially accepted casting practices rather than random fluctuation.

I further refined the analysis by isolating Couple 1 from other couples and standardizing our temporal approach. Separating Couple 1 from the remaining couples revealed that extreme age gaps may be largely driven by star bias, where the most prominent pairings appear to receive age-gap exceptions unavailable to other couples in the film. Shifting from individual-year analysis to decade-level aggregation and gender-specific age lines addressed dataset imbalance and allowed us to visualize the gender age gap, highlighting the limited “shelf life” of actresses compared to the relative agelessness of leading men. Together, these EDA refinements provided the context necessary to justify our machine-learning clusters, showing that age gaps are shaped by the interaction of gender, star power, and historical period.

## 3. Analysis & Experiments

For Milestone 2, I applied two primary unsupervised methods: K-Means Clustering and Silhouette Analysis to investigate the hidden structure of Hollywood's casting history. Here is how these methods address the research question regarding industry power dynamics and social norms.

### 1. K-Means Clustering: Identifying "Casting Regimes"

**Purpose and Impact**: K-Means was used to determine if Hollywood’s history is a single, continuous timeline or a series of distinct "Casting Regimes." By grouping movies based on release year and age difference, this method mathematically identifies the "inflection points" where industry behavior fundamentally changed. This helps address our business question by moving past simple averages and allowing us to see when one set of industry "rules" was replaced by another. I expected it to reveal that the 1940s and the 2010s are not just different years, but entirely different mathematical "neighborhoods" with their own internal logic regarding age gaps.

**Implementation and Adjustments**: I used the Elbow Method to choose the number of clusters, testing values for k ranging from 1 to 8. Initially, I considered using k = 2 for a "Classic vs. Modern" split, but the plot showed a significant bend at k = 3 suggesting a middle "Transition" era was necessary for accuracy. A critical adjustment was the application of StandardScaler to our features; without this, the model would have ignored the age gaps entirely and only clustered by year, because the year values (1930–2022) are numerically much larger than the gaps (0–50).

**Findings and Challenges**: What worked was the clear separation of the "Studio Era," the "Transition Era," and the "Modern Era." A conceptual challenge I encountered was the initial attempt to include actor names or genres in the clustering, this created too much noise and caused the clusters to overlap into a meaningless blur. I learned that for historical trends, focusing on just time and gap allowed the structural boundaries of the industry to emerge clearly.

This taught that while many variables exist, the primary "regimes" are driven by historical time periods.

**2. Silhouette Analysis: Validating Consistency and Finding Outliers**

**Purpose and Impact:** While K-Means gave us the "groups," Silhouette Analysis helped "interrogate" the quality of those groups. It measures how well each movie actually fits into its assigned era. This helps answer our question about "Star Bias" by identifying movies that are "mathematical misfits" films released in a modern year that have an age gap that belongs in the 1940s. I chose this because Hollywood is inconsistent, and I expected to find "negative" values representing movies that defy their historical regime.

**Implementation and Adjustments:** I used the Silhouette Visualizer with our k=3 model. I specifically looked for slivers that cross the 0.0 line into the negatives. I adjusted the focus to the "Transition Era" (the middle cluster) because I suspected that was where the most internal conflict in casting norms would occur. This method was appropriate because it allowed me to look beyond the average and see the individual movies that were "resisting" the trend of their time.

**Findings and Challenges:** I was surprised to find that the "Modern Era" cluster was incredibly stable and consistent (high silhouette scores), while the "Transition Era" (the 1970s and 80s) was very "shaggy" and full of negative scores. This was a breakthrough: it proved that the "Inflection Point" wasn't a clean break, but a messy struggle. I learned that legacy franchises, like the James Bond films, were often the source of these negative scores, they acted as "structural fossils," clinging to old-fashioned age gaps even as the rest of the industry moved forward. This led to abandoning the idea that history moves in a straight line and instead view it as a series of competing power dynamics where some stars or franchises are powerful enough to ignore modern social norms.

## 4. Findings & Interpretations

To summarize our key insights, we discovered that Hollywood’s casting history is defined by three distinct "Historical Regimes" rather than a smooth, gradual timeline. Our clustering analysis (Fig.4) identified a clear break between the "Golden Age" (1930s–1960s), where massive age gaps were the industry standard, and the "Modern Era" (1990s–Present), where casting has stabilized into a much narrower, peer-to-peer norm. The period in between serves as a "Transition Era," marked by significant inconsistency as the industry began to grapple with shifting social expectations. This suggests that for film studios and historians, "progress" isn't a straight line but a series of structural shifts where the rules of what audiences find acceptable are periodically rewritten.

A major driver of these patterns could be the "Star Power" dynamic (Fig.2), which creates a visible divide between Couple 1 and other couples. Our analysis suggested that the most prominent pairings may be granted a "pass" to maintain large age gaps that would be considered outliers among other couples in the cast. Sometimes, while supporting couples generally mirror the social norms of their time, the "Lead Couple" often operates under a different set of rules, with much higher variance and more frequent violations of social benchmarks like the "Half-Your-Age-Plus-Seven" rule (Fig.3). This highlights a tangible business reality in Hollywood: stardom provides institutional leverage that allows specific individuals to ignore the broader cultural shifts toward age equality in romantic pairings.

Finally, our investigation into the "misfits" of the data uncovered that certain legacy franchises act as "structural fossils," resisting change even as the rest of the industry evolves. By identifying movies with negative silhouette scores (Fig.5,6), most notably the James Bond films of the late 70s and 80s, we found that these high-profile properties often clung to "Golden Age" age gaps long after they had become mathematical anomalies for their era. This insight is crucial for understanding brand relevance, it proves that while social norms eventually win out, powerful legacy brands can create "friction" in the data, maintaining outdated casting traditions until a true breaking point or franchise reboot occurs.

## 5. Next Steps

While K-Means revealed clear casting regimes, we have not yet explored the latent space underlying these choices. Our current models focus on time and age, but do not account for hidden film characteristics such as genre, studio, or creative influence that may drive cluster separation. We also plan to apply dimensionality reduction techniques like PCA to compress complex features, such as director tendencies or character archetypes - into a lower-dimensional space to better understand their relationships.

Next, we will use association rule mining to identify recurring casting profiles, such as whether large lead age gaps systematically align with specific genres or directors. We also plan to apply hierarchical clustering to determine whether the identified transition era can be subdivided into finer cultural phases, such as New Hollywood, and to assess whether “misfit” films like the Bond franchise form a distinct legacy cluster or represent random noise.

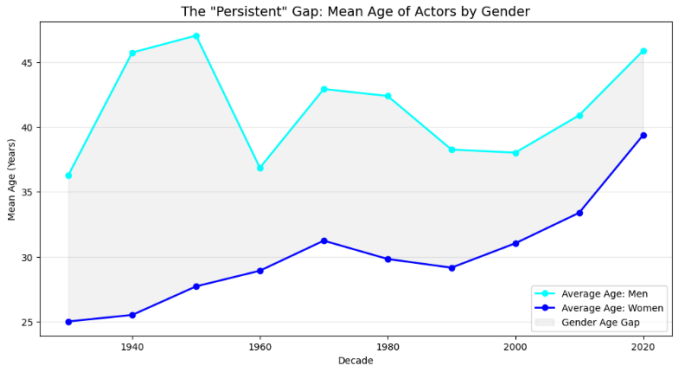
## Appendix

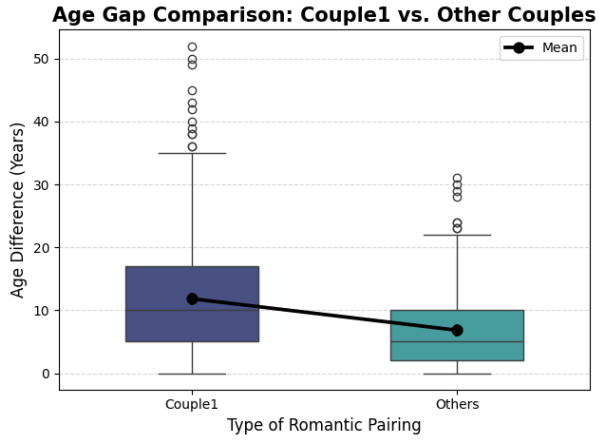
**Shared GitHub Repository - https://github.com/patelparisha/ba820-b1-team10**

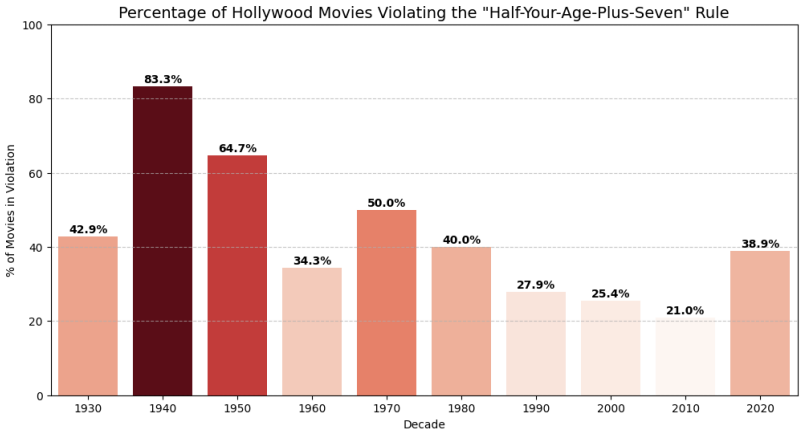
**Individual GitHub Repository - https://github.com/patelparisha/ba820-b1-team10/tree/Milestones/Inchara%20Ashok**

**Google Colab Link - https://colab.research.google.com/drive/1-iNTdOPX76UC2G8uRTuh0sYXsxcy5cEC?usp=chrome_ntp#scrollTo=IYnymIVNwlMJ**

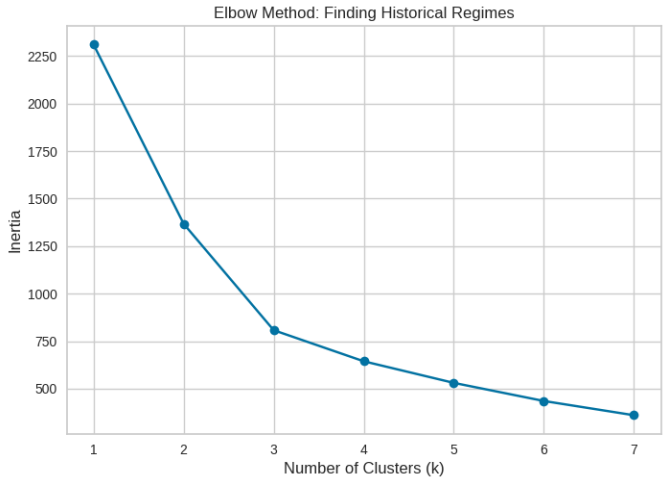
## Supplemental Material

1. 

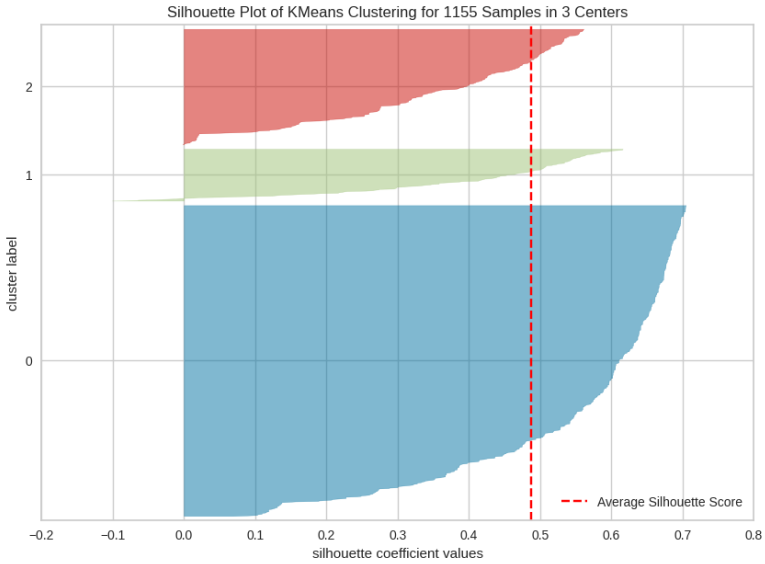

2. 


Percentage of Hollywood Movies Violating the "Half-Your-Age-Plus-Seven" Rule
% of Movies in Violation
Decade
42.9%
83.3%
64.7%
34.3%
50.0%
40.0%
27.9%
25.4%
21.0%
38.9%
0
20
40
60
80
100
1930
1940
1950
1960
1970
1980
1990
2000
2010
2020

3.


Elbow Method: Finding Historical Regimes
Inertia
Number of Clusters (k)
2250
2000
1750
1500
1250
1000
750
500
1
2
3
4
5
6
7

4.


Silhouette Plot of KMeans Clustering for 1155 Samples in 3 Centers
cluster label
silhouette coefficient values
Average Silhouette Score
0
1
2
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

5.

6.

| | movie_name | release_year | age_difference |
|---|---|---|---|
| 99 | Moonraker | 1979 | 24 |
| 134 | For Your Eyes Only | 1981 | 21 |
| 203 | Octopussy | 1983 | 18 |
| 212 | Scarface | 1983 | 18 |
| 425 | Indiana Jones and the Temple of Doom | 1984 | 11 |
| 497 | The Best Little Whorehouse in Texas | 1982 | 10 |
| 541 | Raiders of the Lost Ark | 1981 | 9 |

**External References and Context**


1. Kenrick, D. T. (2018, February 21). *Why Are Older Hollywood Actors Paired With Such Young Women?* Psychology Today. https://www.psychologytoday.com/us/blog/sex-murder-and-the-meaning-of-life/201802/why-are-older-hollywood-actors-paired-with-such
2. Project Fabulous Evee. (2023). *Final project report* (INFO 2950, Spring 2023). Cornell University. https://pages.github.coecis.cornell.edu/info2950-s23/project-fabulous-evee/report.pd


**Process Overview**

The pipeline for this project follows a structured journey from raw data to a deeper understanding of Hollywood's behavior. At the technical level, the process begins with the data moving from individual years to broader historical groups by decade. I used Exploratory Data Analysis (EDA) to find "hidden" signals, such as the clear difference in age gaps between leading stars and the supporting cast. A critical step was standardization, where we adjusted the data so that the year a movie was made and the size of the age gap were treated with equal importance. Conceptually, this "leveled the playing field" so the larger numbers of the years wouldn't drown out the actual age differences we were trying to study.

The core of our machine learning pipeline involves using K-Means Clustering to let the data define its own historical boundaries. By applying the Elbow Method, I mathematically justified dividing Hollywood history into three distinct "Casting Regimes," while Silhouette Analysis helped us find "misfit" movies that didn't follow the rules of their time. This allowed me to prove that certain eras were stable while others were messy periods of conflict between old industry habits and new social rules. The pipeline ends with generating insights, where we connect these mathematical patterns back to real-world examples, like the James Bond franchise. This final step turns abstract numbers into a clear story about how power and legacy can sometimes resist social progress.

**Use of Generative AI Tools**

I used ChatGPT as a supportive tool to help with idea generation, drafting text, and clarifying technical or conceptual questions during the project. All suggestions provided by the AI were critically evaluated, refined, and adapted to fit our own approach and understanding. The final work represents my own analysis, reasoning, and contributions, and AI was used only as an assistant, not as a replacement for our thinking.

**Documentation of AI Usage:**

Link to Chatbot